

# Rofu Thailand rooftop solar and storage

Feasibility results: six  
configurations on the PEA time-of-  
use tariff, sized by optimisation  
and tested on a 20 year pro forma

Prepared by Allotrope Partners  
September 2026



# Summary: solar pays at every roof size; storage is selected

## 1,685-2,106 kWp

usable roof range from the site survey; the economic optimum is larger (2,549 kWp without storage, 2,847 with)

## 367-1,881 kWh

storage selected at both ends of the range (222 kW at the roof-limited size, 507 kW unconstrained)

## 273k-460k USD

year-1 bill savings against a business-as-usual bill of 843,443 USD

## 1.30-1.86 M USD

equity NPV at 11 percent over 20 years; equity IRR 50 to 66 percent on 70 percent debt

## What the analysis found

- Every roof case is strongly positive: equity IRR above 60 percent and simple equity payback under two years, driven by a 500 USD/kWp installed cost that sits below published Thai benchmarks.
- The roof, not the economics, sets the size. Each square metre confirmed adds value up to about 2.5 to 2.8 MWp.
- The site cannot export, so 9 to 19 percent of production is curtailed depending on size; storage recovers part of it and clips the on-peak demand charge.
- The battery is replaced in full in year 10, the final loan year. Case 6 shows a one-year coverage dip that a reserve or a longer tenor removes.

## Recommended path

- Base case: the roof-limited array with storage (case 5): 1,685 kWp, 222 kW / 367 kWh, 919,716 USD capex, NPV 1.35 M USD.
- Upside: the unconstrained array with storage (case 6) if the measured roof allows it: 2,847 kWp, NPV 1.86 M USD.
- Next: measure the roof, obtain an EPC quotation against 500 USD/kWp, decide on grid export, supply monthly kVAR data.

## Question

How much rooftop solar, with or without battery storage, should Rofu Thailand install under its current PEA tariff, and what does the investment return over 20 years?

## Method

- Dispatch and sizing: REopt optimisation at 15 minute resolution, one full year of metered load, PVWatts irradiance for Phimai, no grid export.
- Economics: 20 year pro forma, direct ownership by the factory, 70 percent debt, Thai CIT and depreciation.
- Six cases: three roof sizes, one unconstrained size, and the two storage variants.

## How the battery conclusion was tested

1

### Initial price

300 USD/kW + 250 USD/kWh (675 USD/kW installed for 1.5 h)

**Storage rejected by the optimiser**

2

### Keen quotation

100 USD/kW + 150 USD/kWh (325 USD/kW installed); year-10 replacement at 70 percent of install

**Storage selected: 429 kWh roof-limited, 2,378 kWh unconstrained**

3

### Conservative replacement

Same prices; the whole battery system replaced in year 10 at 100 percent of install cost

**Storage still selected: 367 kWh and 1,881 kWh. The conclusion has held at three price points**

# The facility: Rofu Thailand, Phimai

## Supply

**Utility:** Provincial Electricity Authority, Phimai branch

**Tariff:** Schedule 4.2 Large General Service, time of use, 22-33 kV, rate code 4224

**Connection:** one grid connection, four transformers, 3,230 kVA installed (500 + 500 + 1,600 + 630)

**Location:** Phimai district, Nakhon Ratchasima

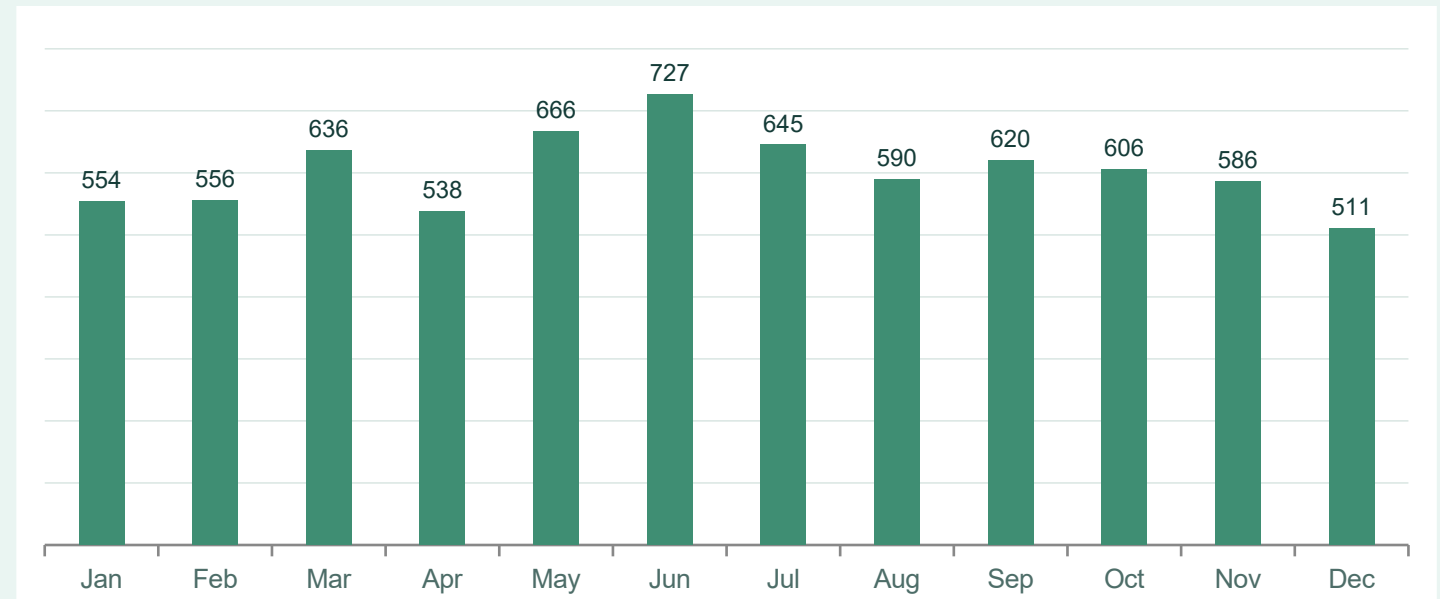
## Roof

**Survey basis:** five roofs, 108 m long, widths between 24 m and 30 m; pitch not recorded

**Usable estimate:** 65 percent of area at 0.20 kW per m<sup>2</sup> gives 1,685 to 2,106 kWp

**Status:** the widths are the uncertain term; the roof has not been measured for this study

## Metered electricity use, 2025 (MWh per month)



**7.24 GWh**

annual consumption

**1.21-1.38 MW**

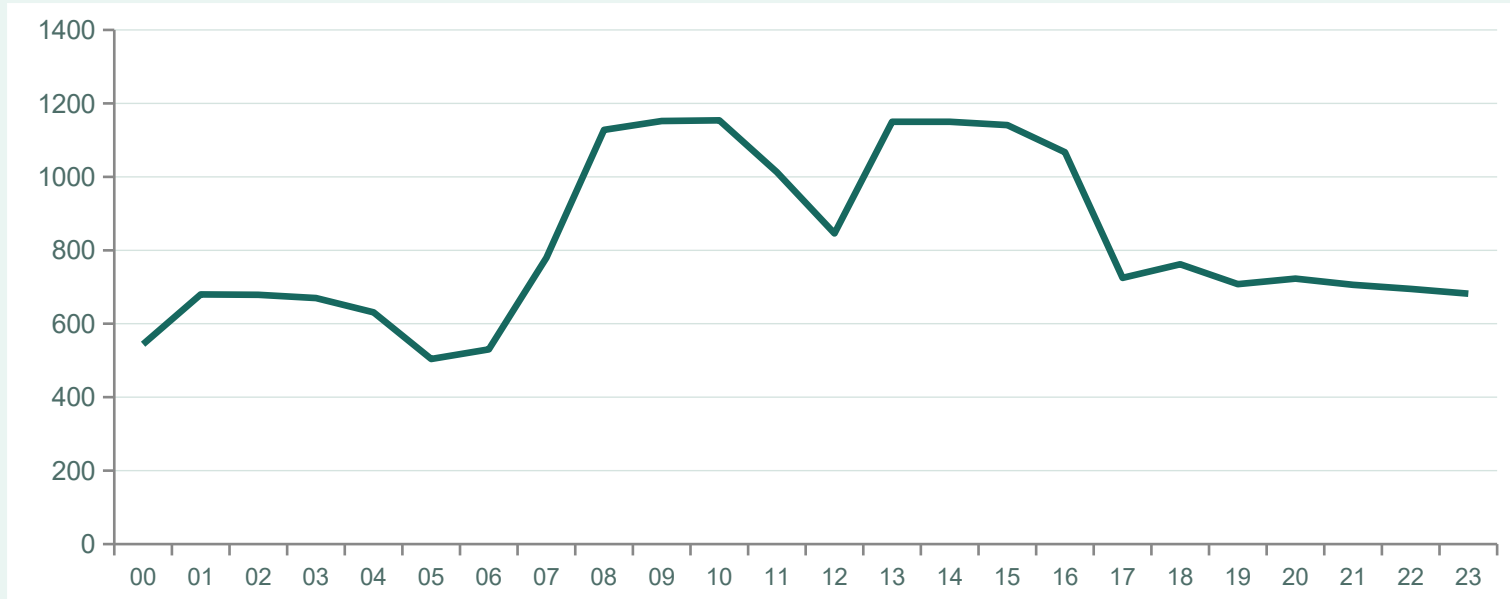
monthly on-peak  
maximum demand

**843,443 USD**

business-as-usual bill per  
year (THB 27.4 million)

# Daily load against the tariff clock and the solar day

## Average load by hour, 2025 (kW)



### Weekday tariff

off-peak 2.6037

peak 4.1839 THB/kWh + monthly demand charge

off

### Solar output

generation 06:00 to 18:00, peak 11:00 to 14:00

00:00

06:00

09:00

12:00

18:00

22:00

24:00

*Weekends and Thai public holidays are off-peak all day.*

## What this means for the design

- Solar displaces the 4.18 THB peak energy from 09:00 to 18:00 on weekdays, the most valuable kWh on the bill.
- The 08:00 to 16:00 load plateau of about 1.15 MW is above the roof-limited array's average output, so most weekday solar is used on site.
- The demand charge is set by one interval a month; PV alone cannot guarantee a lower peak, storage can.
- Weekends and holidays are off-peak all day and the load is lower: that is where curtailment happens without export.

# The PEA tariff: five building blocks, one monthly peak

## Peak energy

**4.1839 THB/kWh**

weekdays 09:00-22:00

## Off-peak energy

**2.6037 THB/kWh**

nights, weekends, holidays

## On-peak demand

**132.93 THB/kW a month**

highest on-peak interval

## Ft adjustment

**0.10-0.37 THB/kWh**

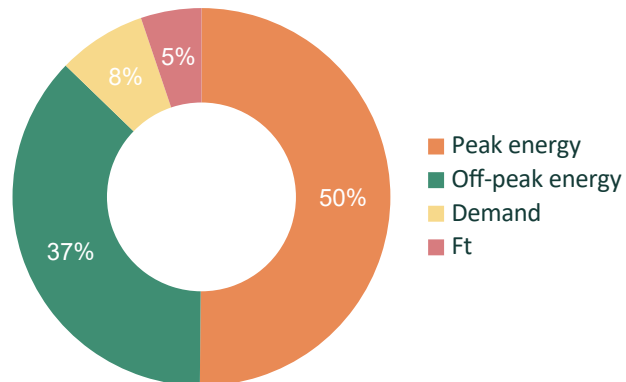
all kWh; reset every four months

## Service and VAT

**312.24 THB/month**

then 7 percent VAT on the subtotal

## June 2025 invoice: THB 2.58 million



**Pre-VAT subtotal THB 2,407,767**

**631,580 kWh** billed in the month

**1,368 kW** on-peak maximum demand, the single interval that sets THB 181,848 of demand charge

**Ft 0.1972 THB/kWh** on every kWh in that period

**Energy is 87 percent of the bill;** demand 8 percent, Ft 5 percent; the service charge is negligible

## Why the structure matters

- Solar earns the peak rate on weekdays and the off-peak rate on weekends and holidays; the blended value of a solar kWh is well below the peak rate.
- One monthly interval sets the demand charge, so a cloudy quarter hour can wipe out a month of peak shaving unless a battery covers it.
- Ft moves every four months and is outside the site's control; the model carries 3 percent a year tariff escalation and does not forecast Ft.

# Assumptions (1): technical and cost inputs

| Item                                                  | Value                                              | Basis                                                                                                            |
|-------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| PV installed cost                                     | 500 USD/kWp                                        | Keen quotation; below published Thai benchmarks (see slide 14)                                                   |
| Battery installed cost                                | 100 USD/kW + 150 USD/kWh                           | Keen quotation; 325 USD/kW installed for a 1.5 hour system                                                       |
| Battery minimum duration                              | 1.5 hours                                          | provisional                                                                                                      |
| PV operating cost                                     | 8.00 USD/kWp per year                              | supplied at 7.50 (1.5 percent of capex); optimiser rounds to whole dollars                                       |
| Battery operating cost                                | 1 percent of install cost per year                 | optimiser convention                                                                                             |
| Insurance                                             | 0.5 percent of capex per year                      | provisional                                                                                                      |
| Specific yield, year 1                                | 1,498 kWh/kWp                                      | PVWatts irradiance for Phimai: 2,049 kWh/m <sup>2</sup> plane of array; performance ratio 0.731; tilt 15 degrees |
| PV degradation                                        | 0.5 percent per year                               | optimiser default                                                                                                |
| Battery replacement                                   | year 10, 100 percent of install cost, whole system | conservative case; capitalised and depreciated over 5 years                                                      |
| PV inverter replacement                               | year 11, 10 percent of PV capex                    | Allotrope convention, provisional                                                                                |
| Grid export                                           | none                                               | strict no-export site; surplus is curtailed                                                                      |
| Grid connection, power factor mitigation, permit fees | excluded from capex                                | to be added when quoted                                                                                          |

# Assumptions (2): financing, tax and scope

## Financing and tax

|                             |                                                            |
|-----------------------------|------------------------------------------------------------|
| <b>Project life</b>         | 20 years                                                   |
| <b>Discount rate</b>        | 11 percent, equity NPV basis                               |
| <b>Debt</b>                 | 70 percent of capex, 6.5 percent, 10 years, level payments |
| <b>Exchange rate</b>        | 32.5 THB per USD, planning rate                            |
| <b>Corporate income tax</b> | 20 percent flat, no incentive regime                       |
| <b>Depreciation</b>         | 5 year machinery life, PV and battery alike                |
| <b>Tariff escalation</b>    | 3 percent a year on the PEA bill (provisional)             |
| <b>O&amp;M escalation</b>   | 3 percent a year                                           |
| <b>VAT</b>                  | 7 percent on the bill                                      |

## Ownership structure

Direct ownership: the factory invests in its own roof and keeps the whole avoided PEA bill. No ESCO discount, no third-party developer margin, no new-project tax incentive. Returns are equity returns after debt service and CIT.

## Emissions and scope

- Avoided emissions: avoided grid import multiplied by 0.4750 kg CO<sub>2</sub>e per kWh (Thailand Greenhouse Gas Management Organization, 2022-2024 grid mix).
- Excluded at the client's direction: grid connection cost, power factor mitigation (no kVAR data supplied), permit and licence fees (no published schedule).
- Fifteen inputs are provisional and marked as such in the workbooks; the ones that move the answer are the roof area, the debt terms and the insurance rate (Appendix A).

*All results are before grid connection, permitting and power-factor costs; add them when quoted.*

# Six configurations were optimised

## 1 Roof, low

PV capped at 1,685 kWp (24 m roof width)  
**no storage**

## 2 Roof, mid

PV capped at 1,895 kWp (27 m)  
**no storage**

## 3 Roof, high

PV capped at 2,106 kWp (30 m)  
**no storage**

## 4 No roof limit

PV free up to the 3,230 kVA connection  
**no storage: the economic optimum**

## 5 Roof, low, with storage

PV capped at 1,685 kWp  
**storage sized by the optimiser**

## 6 No roof limit, with storage

PV free  
**storage sized by the optimiser**

### How the roof estimate is built

5 roofs

x 108 m

x 24 to 30 m

x 65% usable

x 0.20 kW/m<sup>2</sup>

**= 1,685 to 2,106 kWp**

The widths are the uncertain term and the pitch was not recorded. The optimiser wants 2,549 kWp without storage and 2,847 kWp with it, so the roof binds across the whole plausible range.

# Results: the six cases side by side

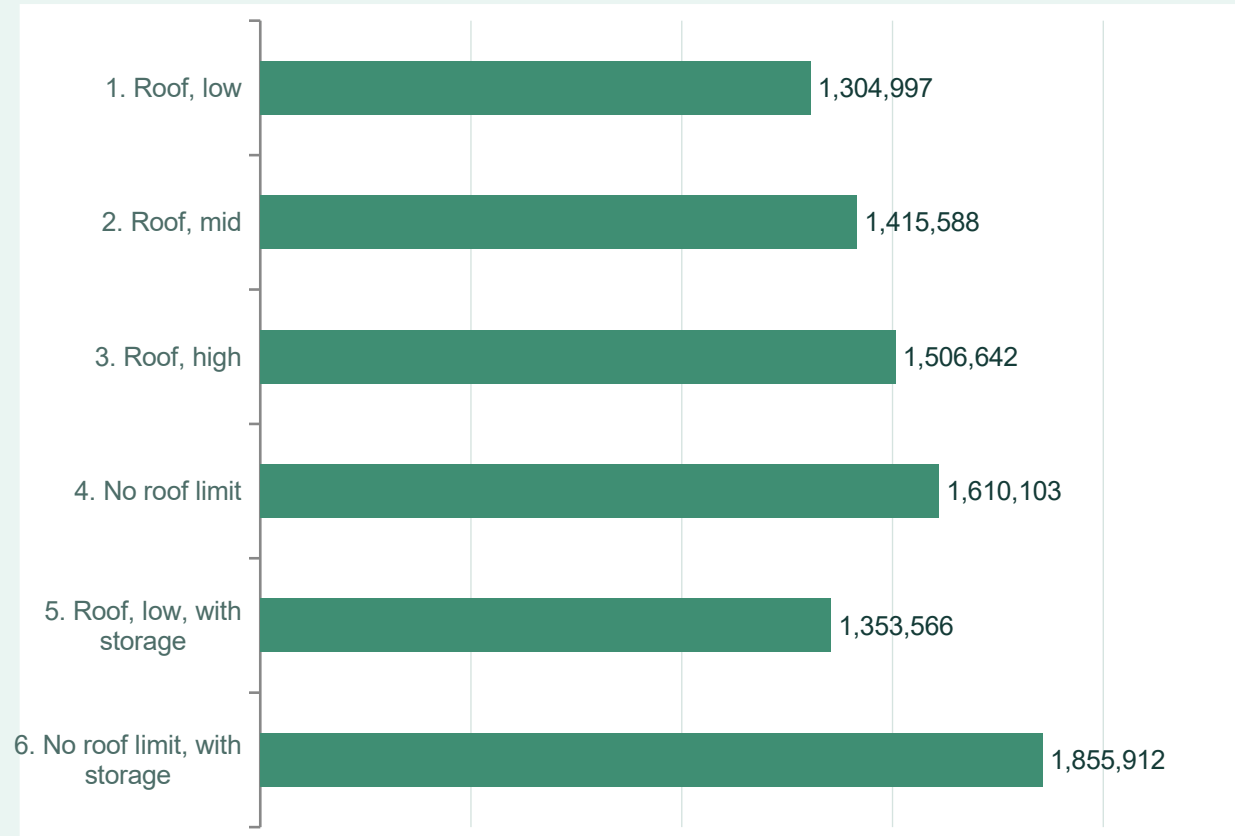
| Case                                  | PV (kWp)     | Storage                   | Capex (USD)      | Year-1 savings (USD) | Grid offset  | tCO2e / yr   | Equity IRR   | Equity NPV (USD) |
|---------------------------------------|--------------|---------------------------|------------------|----------------------|--------------|--------------|--------------|------------------|
| 1. Roof, low                          | 1,685        | none                      | 842,500          | 272,717              | 30.8%        | 1,096        | 66.4%        | 1,304,997        |
| 2. Roof, mid                          | 1,895        | none                      | 947,500          | 299,784              | 33.8%        | 1,205        | 64.5%        | 1,415,588        |
| 3. Roof, high                         | 2,106        | none                      | 1,053,000        | 324,301              | 36.6%        | 1,304        | 62.2%        | 1,506,642        |
| 4. No roof limit                      | 2,549        | none                      | 1,274,321        | 364,048              | 41.1%        | 1,465        | 56.1%        | 1,610,103        |
| <b>5. Roof, low, with storage</b>     | <b>1,685</b> | <b>222 kW / 367 kWh</b>   | <b>919,716</b>   | <b>289,920</b>       | <b>32.1%</b> | <b>1,116</b> | <b>64.2%</b> | <b>1,353,566</b> |
| <b>6. No roof limit, with storage</b> | <b>2,847</b> | <b>507 kW / 1,881 kWh</b> | <b>1,756,407</b> | <b>460,458</b>       | <b>50.4%</b> | <b>1,740</b> | <b>49.6%</b> | <b>1,855,912</b> |

## Reading the table

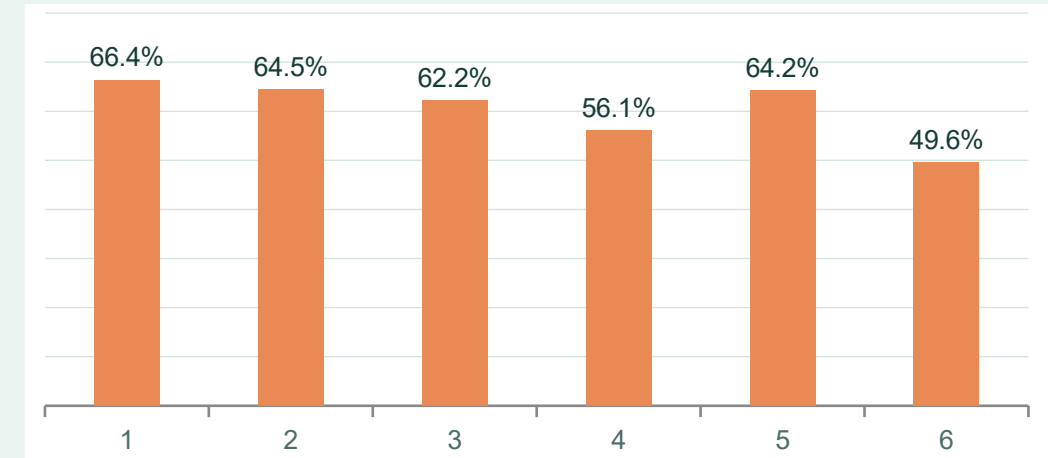
- Every case returns its equity in under two years at the 500 USD/kWp price; the differences between cases are in scale, not in whether the investment works.
- Storage adds 77,000 USD of capex at the roof-limited size and 17,000 USD of year-1 savings, lifting NPV by 49,000 USD; unconstrained, it adds 1.9 MWh and lifts NPV by 246,000 USD.
- Equity IRR falls with size because each added kilowatt is used less on site; NPV, not IRR, is the figure to size on.

# Bigger systems return more in total, less per dollar

## Equity NPV (USD, 20 years at 11 percent)



## Equity IRR



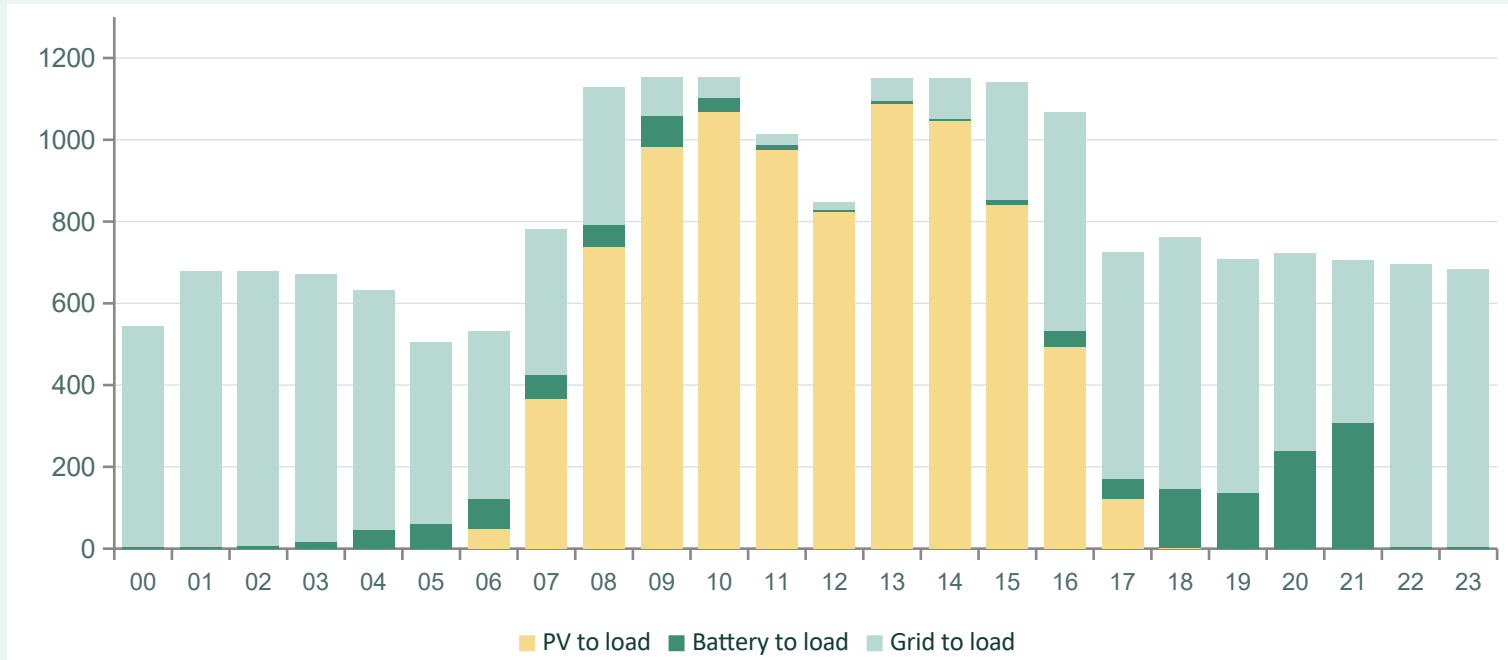
### Capex and equity by case (USD)

- 1: capex 842,500, equity 252,750, debt 589,750
- 2: capex 947,500, equity 284,250, debt 663,250
- 3: capex 1,053,000, equity 315,900, debt 737,100
- 4: capex 1,274,321, equity 382,296, debt 892,025
- 5: capex 919,716, equity 275,915, debt 643,801
- 6: capex 1,756,407, equity 526,922, debt 1,229,485

*Simple equity payback runs from 1.5 years (case 1) to 2.0 years (case 6). These are consequences of the 500 USD/kWp input; see slide 14.*

# What storage buys: recovered curtailment and a lower monthly peak

Case 6, average day: who serves the load (kW by hour)



Late morning the array also charges the battery (up to 340 kW) and still curtails up to 400 kW; the battery gives it back from 18:00 to 22:00, inside the peak window.

**529 MWh / yr**

delivered by the case 6 battery to the load, mostly in the evening peak hours

**8.6% to 6.5%**

curtailment at the roof-limited array without and with storage (cases 1 and 5): energy that would otherwise be lost

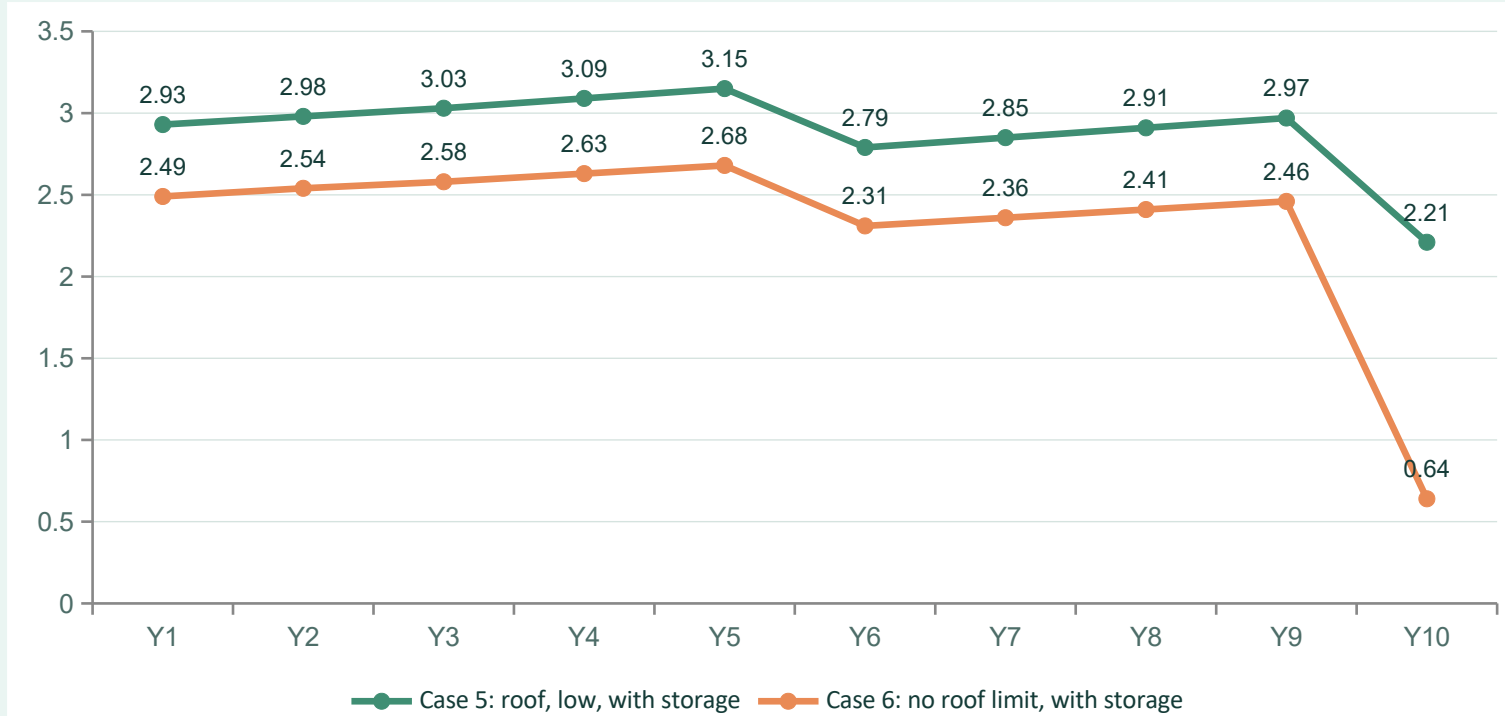
**32,470 USD / yr**

case 6 demand charge savings; the battery covers the interval that would set the monthly peak

The conclusion held at three battery price points, so it rests on the tariff shape and the no-export rule, not on one quotation.

# The year-10 battery replacement meets the final loan year

## Debt service coverage ratio, years 1 to 10



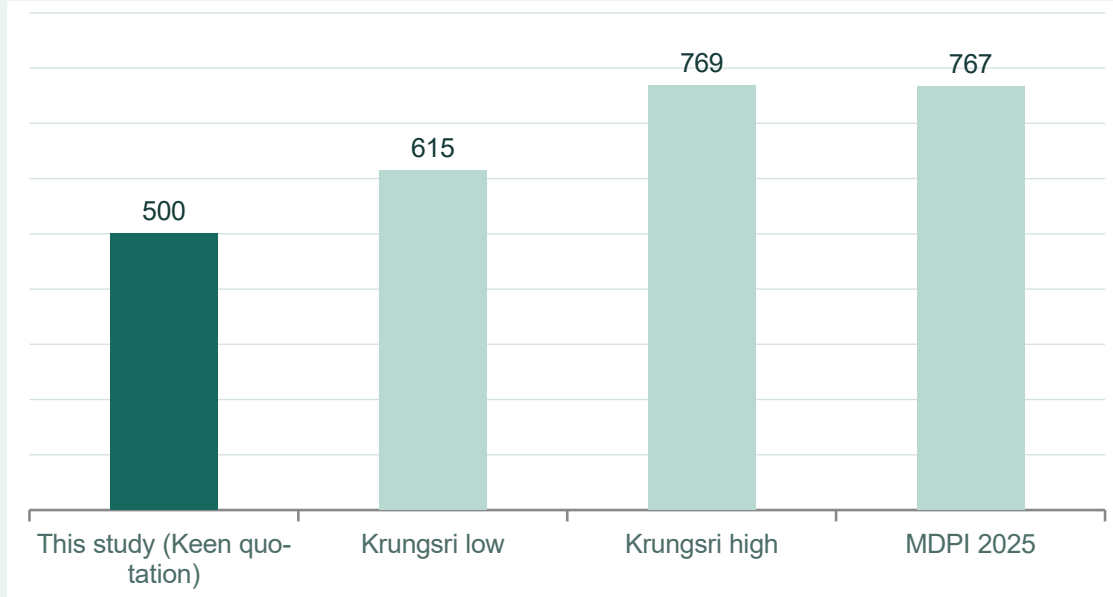
Average coverage over the loan: case 5 2.89, case 6 2.31. PV inverter replacement (10 percent of PV capex) follows in year 11, after the loan.

## A structuring point

- Case 6, year 10: cash available for debt service 109,516 USD against 171,027 USD of debt service after a 332,891 USD battery replacement.
- A lender will expect the replacement funded from a reserve built in years 1 to 9, or a tenor set so the replacement falls after the final payment. Either fix is routine.
- The minimum coverage figure on the case 6 workbook should be read with this in mind.
- The optimiser's own degradation model puts the case 6 battery at 96 percent capacity after 20 years; the year-10 full replacement is conservative by a wide margin.

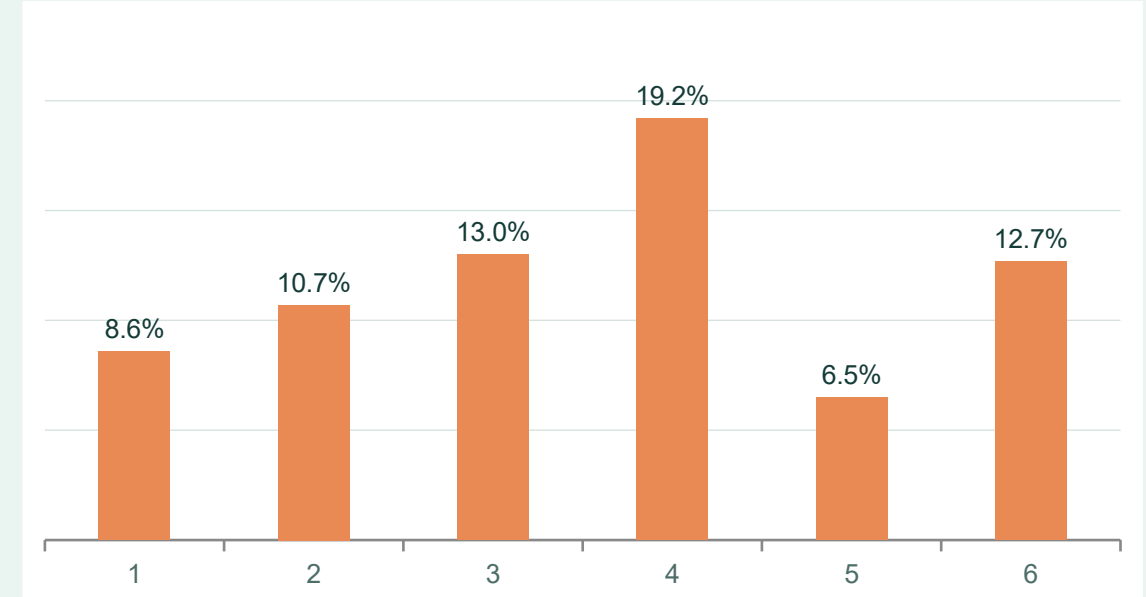
# Two inputs move the answer: capital cost and export

## Installed PV cost, USD per kWp



The 500 USD/kWp price is used as given. It is why simple equity payback is 1.5 years; a quotation at 650 USD/kWp would change the picture materially. Confirming the price is worth more than refining anything else in the analysis.

## Share of production curtailed (no export)



At the unconstrained optimum nearly one fifth of generation is discarded. An export price, if PEA or a third party offers one, would raise both the value of the larger systems and the optimal size; the question is left open rather than priced by assumption.

Cases: 1 roof low, 2 roof mid, 3 roof high, 4 no roof limit, 5 roof low with storage, 6 no roof limit with storage.

# Recommendation: build to the roof, with storage

## Base case: case 5

**1,685 kWp PV with 222 kW / 367 kWh storage**

Capex 919,716 USD, equity 275,915 USD

Year-1 savings 289,920 USD; grid offset 32 percent; 1,116 tCO<sub>2</sub>e a year avoided

Equity NPV 1,353,566 USD; IRR 64.2%; coverage above 2.2 in every year

Fits the smallest roof reading; every metre of roof confirmed beyond it adds value.

## Upside: case 6, if the roof allows

**2,847 kWp PV with 507 kW / 1,881 kWh storage**

Capex 1,756,407 USD, equity 526,922 USD

Year-1 savings 460,458 USD; grid offset 50 percent; 1,740 tCO<sub>2</sub>e a year avoided

Equity NPV 1,855,912 USD; IRR 49.6%; year-10 replacement needs a reserve or a longer tenor

Needs about 35 percent more roof than the most generous survey reading.

## Next steps, in order of value

### 1 Measure the roof

Widths and pitch of the five roofs. The optimum is 2.5 to 2.8 MWp; every confirmed square metre earns.

### 2 Obtain an EPC quotation

Check it against the 500 USD/kWp used here. Every return in this deck scales off that figure.

### 3 Decide on grid export

A price for surplus generation would raise the optimal size; none is assumed today.

### 4 Supply monthly kVAR maxima

Needed to assess power factor; mitigation is currently outside the capital cost.

*Then: ERC generation licence and Aor.6 building modification permit (no published fee schedule); no environmental assessment below 5 MWp.*

# Appendix A: provisional inputs

| Input                       | Value used                                               | Effect if changed                                                          |
|-----------------------------|----------------------------------------------------------|----------------------------------------------------------------------------|
| Usable roof area and PV cap | 8,424 m2, 1,685 kWp (low reading); mid 1,895, high 2,106 | Sets the size of every roof case; the highest-value open item              |
| Debt fraction               | 70 percent of capex                                      | Moves equity IRR and DSCR; NPV to equity changes with leverage             |
| Debt term                   | 10 years                                                 | A longer tenor takes the year-10 replacement past the final payment        |
| Insurance rate              | 0.5 percent of capex per year                            | Recurring cost; second-order at this scale                                 |
| PEA tariff escalation       | 3 percent per year                                       | Raises or lowers every year's savings; Ft is not forecast                  |
| Battery minimum duration    | 1.5 hours                                                | Shapes the kW to kWh ratio the optimiser may choose                        |
| Battery O&M                 | 1 percent of install cost per year                       | Small; offset by the conservative replacement assumption                   |
| PV inverter replacement     | year 11, 10 percent of PV capex                          | Allotrope convention; a different year or share moves one year's cash flow |
| PV tilt                     | 15 degrees                                               | Pitch not recorded; yield sensitivity is small at this latitude            |
| Grid connection cost        | excluded; placeholder pending quotation                  | Add when quoted                                                            |
| Permitting and licence fees | excluded (no published schedule)                         | Add when quoted                                                            |
| Power factor mitigation     | excluded (no kVAR data)                                  | Conditional charge of 56.07 THB per kVAR above the allowance               |

*Counted as fifteen items on the workbook because the roof area and the PV cap are listed separately, as are the PV inverter year and share.*

## Optimisation

- REopt (NREL) v0.57 through Allotrope's REopt API, HiGHS solver.
- 15 minute resolution, one metered year of load (2025).
- PVWatts irradiance for Phimai; tilt 15 degrees; 0.5 percent a year PV degradation.
- Objective: lowest 20 year cost of electricity for the site; no export; storage may charge from grid or PV.

## Pro forma

- 20 year cash flow, direct ownership, 70 percent debt at 6.5 percent over 10 years.
- Thai CIT 20 percent flat; 5 year machinery depreciation; 7 percent VAT on the bill.
- Battery replaced in year 10 at 100 percent of install cost; PV inverter in year 11 at 10 percent of PV capex; both capitalised.
- Every workbook reconciles its live formulas to the engine (audit sheet).

## Emissions and permitting

- Avoided import x 0.4750 kg CO<sub>2</sub>e per kWh (TGO 2022-2024 grid mix, effective 1 January 2026).
- The optimiser's emissions outputs use US datasets and are not used.
- ERC generation licence and Aor.6 building modification permit required; no published fee schedules.
- IEE at 5 MWp, EIA at 10 MWp: not triggered below 3 MWp.

Files: six pro forma workbooks (one per case), the memo of 12 September 2026 and its Vietnamese translation, the tariff structure briefing, and the case inputs, solver payloads and raw results for each case.